Serii Taylor

• Să se determine polinomul Taylor de grad specificat, în punctele indicate, pentru următoarele funcții.

1. $f(x) = -\frac{1}{\sqrt{x+1}}$; $a=0$; $n=3$.

$f'(x) = \left[-(x+1)^{-\frac{1}{2}}\right]' = \frac{1}{2}(x+1)^{-\frac{3}{2}} \Rightarrow$

$\Rightarrow f'(0) = \frac{1}{2}(0+1)^{-\frac{3}{2}} = \frac{1}{2}$

$f''(x) = \frac{1}{2}\cdot\frac{-3}{2}(x+1)^{-\frac{5}{2}} = \frac{-3}{4}(x+1)^{-\frac{5}{2}} \Rightarrow$

$\Rightarrow f''(0) = \frac{-3}{4}(0+1)^{-\frac{5}{2}} = -\frac{3}{4}$

$f'''(x) = \frac{-3}{4}\cdot\frac{-5}{2}(x+1)^{-\frac{7}{2}} = \frac{+15}{8}(x+1)^{-\frac{7}{2}} \Rightarrow$

$\Rightarrow f'''(0) = \frac{15}{8}(0+1)^{-\frac{7}{2}} = \frac{15}{8}$

$\Rightarrow T_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}\cdot x^3 =$

$= -1 + \frac{1}{2}x - \frac{3}{8}x^2 + \frac{5}{16}x^3.$

2. $f(x) = \operatorname{tg} x$; $a=0$; $n=3$

$f'(x) = 1+\operatorname{tg}^2 x \Rightarrow f'(0) = 1+\operatorname{tg}^2 0 = 1$

$f''(x) = 2\operatorname{tg} x(1+\operatorname{tg}^2 x) \Rightarrow f''(0) = 2\cdot 0(1+0) = 0$

$f'''(x) = 2(1+\operatorname{tg}^2 x)^2 + 2\operatorname{tg} x\cdot 2\operatorname{tg} x(1+\operatorname{tg}^2 x) =$

$= 2(1+\operatorname{tg}^2 x)^2 + 4\operatorname{tg}^2 x(1+\operatorname{tg}^2 x) \Rightarrow$

$\Rightarrow f'''(0) = 2(1+0)^2 + 4\cdot 0(1+0) = 2$

$\Rightarrow T_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 =$

$= 0 + 1x + \frac{2}{2}x^2 + \frac{2}{6}x^3 =$

$= x + \frac{1}{3}x^3$

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3. $f(x) = \arcsin x \; ; \; a = 0 ; \; m = 2$

$$f'(x) = \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-\frac{1}{2}} \Rightarrow f'(0) = 1$$

$$f''(x) = -\frac{1}{2}(1-x^2)^{-\frac{3}{2}} \cdot (-2x) \Rightarrow f''(0) = 0$$

$$\Rightarrow T_2(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 =$$

$$= 0 + \frac{1}{1}x + \frac{0}{2}x^2 = x$$

4. $f(x) = x^2 - 2x + 5 \; ; \; a = 0 ; \; m = 3$

$$f'(x) = 2x - 2 \Rightarrow f'(0) = -2$$

$$f''(x) = 2 \Rightarrow f''(0) = 2$$

$$f'''(x) = 0 \Rightarrow f'''(0) = 0$$

$$\Rightarrow T_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 =$$

$$= 5 + (-2)x + 1 \cdot x^2 + 0 \cdot x^3 =$$

$$= x^2 - 2x + 5 = f(x)$$

5. $f(x) = x^{\frac{3}{2}} \; ; \; a = 1 \; ; \; m = 4$

$$f'(x) = \frac{3}{2}x^{\frac{1}{2}} \Rightarrow f'(1) = \frac{3}{2}$$

$$f''(x) = \frac{3}{2} \cdot \frac{1}{2} x^{-\frac{1}{2}} = \frac{3}{4}x^{-\frac{1}{2}} \Rightarrow f''(1) = \frac{3}{4}$$

$$f'''(x) = \frac{3}{4}\left(-\frac{1}{2}\right)x^{-\frac{3}{2}} = \frac{-3}{8}x^{-\frac{3}{2}} \Rightarrow f'''(1) = \frac{-3}{8}$$

$$f^{(IV)}(x) = \frac{-3}{8}\left(\frac{-3}{2}\right)x^{-\frac{5}{2}} = \frac{9}{16}x^{-\frac{5}{2}} \Rightarrow f^{(IV)}(1) = \frac{9}{16}$$

$$\Rightarrow T_4(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{(IV)}(1)}{4!}(x-1)^4 =$$

$$= 1 + \frac{3}{2}(x-1) + \left(\frac{-3}{16}\right)(x-1)^2 + \left(\frac{-1}{16}\right)(x-1)^3 + \frac{3}{128}(x-1)^4 =$$

$$= \frac{3}{128}(x-1)^4 - \frac{1}{16}(x-1)^3 - \frac{3}{16}(x-1)^2 + \frac{3}{2}(x-1) + 1$$

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6. $f(x) = \sin x$; $a = \frac{\pi}{2}$; $n = 4$

$f'(x) = \cos x \Rightarrow f'\left(\frac{\pi}{2}\right) = 0$

$f''(x) = -\sin x \Rightarrow f''\left(\frac{\pi}{2}\right) = -1$

$f'''(x) = -\cos x \Rightarrow f'''\left(\frac{\pi}{2}\right) = 0$

$f^{(IV)}(x) = \sin x \Rightarrow f^{(IV)}\left(\frac{\pi}{2}\right) = 1$

$$\Rightarrow T_4(x) = f\left(\frac{\pi}{2}\right) + \frac{f'\left(\frac{\pi}{2}\right)}{1!}\left(x-\frac{\pi}{2}\right) + \frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x-\frac{\pi}{2}\right)^2 + \frac{f'''\left(\frac{\pi}{2}\right)}{3!}\left(x-\frac{\pi}{2}\right)^3 + \frac{f^{(IV)}\left(\frac{\pi}{2}\right)}{4!}\left(x-\frac{\pi}{2}\right)^4$$

$$T_4(x) = 1 + \frac{0}{1}\left(x-\frac{\pi}{2}\right) + \frac{-1}{2}\left(x-\frac{\pi}{2}\right)^2 + \frac{0}{6}\left(x-\frac{\pi}{2}\right)^3 + \frac{1}{24}\left(x-\frac{\pi}{2}\right)^4 =$$

$$= \frac{1}{24}\left(x-\frac{\pi}{2}\right)^4 - \frac{1}{2}\left(x-\frac{\pi}{2}\right)^2 + 1$$

• Determinați reprezentarea în serie Taylor și multimea de convergență a acestei serii pentru următoarele funcții (se consideră $a=0$ și n arbitrar) folosind formulele derivatelor de ordin n și formula seriei lui Newton:

$$(1+y)^\alpha = 1 + \sum_{n\geq 1} \frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1)}{n!} y^n.$$

1. $f(x) = \frac{1}{1-x}$

$$f^{(n)}(x) = \frac{(-1)^n \cdot n!(-1)^n}{(1-x)^n} \Rightarrow f^{(n)}(0) = \frac{(-1)^{2n}\cdot n!}{1^n} = n!$$

$$f(x) = \frac{1}{1-x} = (1-x)^{-1} = 1 + \sum_{n\geq 1} \frac{(-1)(-2)(-3)\dots(-n)}{n!}(-x)^n =$$

$$= 1 - 1 + \sum_{n\geq 0} \frac{(-1)^n n!}{n!}(-1)^n \cdot x^n = \sum_{n\geq 1} x^n \Rightarrow a_n = 1$$

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{1}{1} = 1 \Rightarrow R = \frac{1}{L} = 1$$

pt $x=1$: $\sum_{n\geq 0} 1$ divergentă

pt $x=-1$: $\sum_{n\geq 0}(-1)^n$ divergentă (nu există $\lim_{n\to\infty}(-1)^n$)

$\Rightarrow$ M.C. : $(-1,1)$.

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2 $f(x) = \ln(x+1)$

$f'(x) = \frac{1}{x+1} \Rightarrow f^{(n)}(x) = \left(f'(x)\right)^{(n-1)} = \frac{(-1)^n (n-1)! \cdot 1^n}{(x+1)^n} \Rightarrow$

$\Rightarrow f^{(n)}(0) = \frac{(-1)^n (n-1)!}{1^n} = (-1)^n (n-1)!$

$f(x) = f(0) + \sum_{n\geq 1} (-1)^{n-1}(n-1)! \frac{x^n}{n!} = 0 + \sum_{n\geq 1} \frac{(-1)^{n-1}}{n} \cdot x^n =$

$= \sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n \; ; \; a_n = \frac{(-1)^{n-1}}{n}$

$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^n}{n+1} \cdot \frac{n}{(-1)^{n-1}} \right| = \lim_{n\to\infty} \frac{n}{n+1} = 1$

$\Rightarrow R = \frac{1}{L} = 1$

pt $x = 1$: $\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} = -\sum_{n\geq 1} \frac{(-1)^n}{n}$; $x_n = \frac{1}{n}$ și

aplicăm criteriul Leibnitz:

$\frac{x_{n+1}}{x_n} = \frac{n}{n+1} < 1 \Rightarrow x_{n+1} < x_n \Rightarrow x_n \searrow 0$ (1)

$\lim_{n\to\infty} \frac{1}{n} = 0$ (2)

(1)(2) $\Rightarrow \sum_{n\geq 1} (-1)^n x_n$ convergentă

pt $x = -1$: $\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} \cdot (-1)^n = \sum_{n\geq 1} \frac{(-1)^{2n-1}}{n} = -\sum_{n\geq 1} \frac{1}{n}$ divergentă

(serie armonică generalizată cu $\alpha = 1$)

$\Rightarrow$ M.C. : $(-1, 1]$

3 $f(x) = e^x$

$f^{(n)}(x) = e^x \Rightarrow f^{(n)}(0) = 1$

$f(x) = \sum_{n\geq 0} \frac{1}{n!} x^n \; ; \; a_n = \frac{1}{n!}$

$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{n!}{(n+1)!} = \lim_{n\to\infty} \frac{1}{n+1} = 0 \Rightarrow$

$\Rightarrow R = \frac{1}{L} = +\infty \Rightarrow$ M.C. : $\mathbb{R}$

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4. $f(x) = \cos x$

$f^{(m)}(x) = \cos\left(x + \frac{m\pi}{2}\right) \Rightarrow f^{(m)}(0) = \cos\left(\frac{m\pi}{2}\right)$

$\Rightarrow f(x) = \sum_{m\geq 0} \frac{\cos\left(\frac{m\pi}{2}\right)}{m!} \cdot x^m = \frac{\cos 0}{0!} \cdot x^0 + \frac{\cos\frac{\pi}{2}}{1!} \cdot x +$

$+ \frac{\cos\pi}{2!} x^2 + \frac{\cos\frac{3\pi}{2}}{3!} x^3 + \ldots = 1 + 0 + \frac{-1}{2!} x^2 + 0 + \frac{1}{4!} x^4 +$

$+ 0 + \frac{-1}{6!} x^6 + \ldots = \sum_{n\geq 0} (-1)^n \cdot \frac{1}{(2n)!} x^{2n}$

Notăm: $a_n = \frac{(-1)^n}{(2n)!}$ și $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{(-1)^n}\right| =$

$= \lim_{n\to\infty} \frac{1}{(2n+1)(2n+2)} = 0 \Rightarrow R = \frac{1}{L} = +\infty \Rightarrow$ M.C. : $\mathbb{R}$

5. $f(x) = \sin x$

$f^{(n)}(x) = \sin\left(x + \frac{n\pi}{2}\right) \Rightarrow f^{(n)}(0) = \sin\left(\frac{n\pi}{2}\right)$

$\Rightarrow f(x) = \sum_{n\geq 0} \frac{\sin\left(\frac{n\pi}{2}\right)}{n!} \cdot x^n = \frac{\sin 0}{0!} \cdot x^0 + \frac{\sin\frac{\pi}{2}}{1!} \cdot x +$

$+ \frac{\sin\pi}{2!} \cdot x^2 + \frac{\sin\frac{3\pi}{2}}{3!} \cdot x^3 + \frac{\sin 2\pi}{4!} \cdot x^4 + \ldots = \ldots$

$= 0 + \frac{1}{1!} \cdot x + 0 + \frac{-1}{3!} \cdot x^3 + 0 + \ldots =$

$= \sum_{n\geq 0} (-1)^n \frac{1}{(2n+1)!} x^{2n+1} \Rightarrow a_n = \frac{(-1)^n}{(2n+1)!}$

$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-1)^{n+1}}{(2n+3)!} \cdot \frac{(2n+1)!}{(-1)^n}\right| =$

$= \lim_{n\to\infty} \frac{1}{(2n+2)(2n+3)} = 0 \Rightarrow R = \frac{1}{L} = +\infty \Rightarrow$

$\Rightarrow$ M.C. : $\mathbb{R}$

6. $f(x) = \operatorname{arctg} x$

$f'(x) = \frac{1}{1+x^2} = (1+x^2)^{-1} = 1 + \sum_{n\geq 1} \frac{(-1)(-2)\ldots(-n)}{n!} \cdot x^{2n} =$

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$= 1 + \sum_{n\geq 1} (-1)^n \cdot x^{2n} = 1 - 1 + \sum_{n\geq 0} (-1)^n x^{2n} = \sum_{n\geq 0} (-1)^n x^{2n} \mid \int dx$

$\Rightarrow f(x) = \sum_{n\geq 0} (-1)^n \frac{1}{2n+1} x^{2n+1} + C$

$x = 0 \Rightarrow f(0) = \operatorname{arctg} 0 = 0$

$f(0) = \sum_{n\geq 0} (-1)^n \cdot \frac{1}{2n+1} \cdot 0 + C \Bigg\} \Rightarrow C = 0$

$\Rightarrow f(x) = \sum_{n\geq 0} (-1)^n \cdot \frac{1}{2n+1} \cdot x^{2n+1}$; $a_n = \frac{(-1)^n}{2n+1}$

$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^{n+1}}{2n+3} \cdot \frac{2n+1}{(-1)^n} \right| =$

$= \lim_{n\to\infty} \frac{2n+1}{2n+3} = 1 \Rightarrow R = \frac{1}{L} = 1$

pt $x = 1 \Rightarrow \sum_{n\geq 0} (-1)^n \cdot \frac{1}{2n+1}$. Notăm: $x_n = \frac{1}{2n+1}$ și folosim criteriul Leibnitz:

$\frac{x_{n+1}}{x_n} = \frac{2n+1}{2n+3} < 1 \Rightarrow x_{n+1} < x_n \Leftrightarrow x_n \downarrow_s$ (1)

$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{2n+1} = 0$ (2)

$\xRightarrow{(1)(2)} \sum_{n\geq 0} a_n$ convergentă

pt $x = -1 \Rightarrow \sum_{n\geq 0} (-1)^n \cdot \frac{1}{2n+1} (-1)^{2n+1} = \sum_{n\geq 0} \frac{(-1)^{3n+1}}{2n+1}$

convergentă (criteriul lui Leibnitz)

$\Rightarrow$ M.C. : $[-1, 1]$

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• Dezvoltați următoarele funcții în serie de puteri ale lui $(x-a)$ și determinați mulțimea de convergență a seriei obținute.

1. $f(x) = x^3 - 2x^2 - 5x - 2 \; ; a = -4$

$$f(x) = \sum_{n\geq 0} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$f(-4) = -64 - 32 + 20 - 2 = -78$

$f'(x) = 3x^2 - 4x - 5 \Rightarrow f'(-4) = 48 + 16 - 5 = 59$

$f''(x) = 6x - 4 \Rightarrow f''(-4) = -24 - 4 = -28$

$f'''(x) = 6 \Rightarrow f'''(-4) = 6$

pt $k \geq 4 \Rightarrow f^{(k)}(x) = 0$

$$\Rightarrow f(x) = f(-4) + \frac{f'(-4)}{1!}(x+4) + \frac{f''(-4)}{2!}(x+4)^2 + \frac{f'''(-4)}{3!}(x+4)^3$$

$$= -78 + 59(x+4) - 14(x+4)^2 + (x+4)^3$$

2. $f(x) = \ln x \; ; a = 1$

$$f(x) = \sum_{n\geq 0} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$f'(x) = \frac{1}{x} \Rightarrow f^{(n)}(x) = \left(\frac{1}{x}\right)^{(n-1)} = \frac{(-1)^{n-1}(n-1)!}{x^n}$$

$$f^{(n)}(1) = (-1)^{n-1}(n-1)!$$

$$f(x) = \underbrace{f(1)}_{0} + \sum_{n\geq 1} \frac{(-1)^n (n-1)!}{n!}\cdot(x-1)^n = \sum_{n\geq 1} \frac{(-1)^n}{n}\cdot(x-1)^n$$

Notăm: $x - 1 = y \Rightarrow \sum_{n\geq 1} \frac{(-1)^n}{n} y^n$. Notăm: $\frac{(-1)^n}{n} = a_n$

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-1)^{n+1}}{n+1}\cdot\frac{n}{(-1)^n}\right| = \lim_{n\to\infty} \frac{n}{n+1} = 1$$

$$\Rightarrow R = \frac{1}{L} = 1$$

pt $y = 1 \Rightarrow \sum_{n\geq 1} \frac{(-1)^n}{n}$. Notăm: $u_n = \frac{1}{n}$ și folosim criteriul Leibnitz:

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$\frac{u_{n+1}}{u_n} = \frac{n}{n+1} < 1 \Rightarrow u_{n+1} < u_n \Rightarrow u_n \downarrow_s (1)$

$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n} = 0 (2)$

$\xrightarrow{(1)(2)} \sum_{n\geq 1} a_n$ convergentă

pt $y=-1$: $\sum_{n\geq 1} \frac{(-1)^n}{n} \cdot (-1)^n = \sum_{n\geq 1} \frac{1}{n}$ divergentă (serie armonică generalizată cu $\alpha = 1$)

$\Rightarrow$ M.C. în $y = (-1, 1] \Rightarrow$ M.C. în $x = (0, 2]$

3. $f(x) = \frac{1}{x^2}$; $a = -1$

$f(x) = \sum_{n\geq 0} \frac{f^{(n)}(a)}{n!} \cdot (x-a)^n$

$\int f(x)dx = \frac{1}{-x} \Rightarrow f^{(n)}(x) = \left(\int f(x)dx\right)^{(n+1)} = \frac{(-1)^{n+1}(n+1)!\,(-1)^{n+1}}{(-x)^{n+2}} =$

$= \frac{(n+1)!}{x^{n+2}}$

$\Rightarrow f^{(n)}(-1) = \frac{(n+1)!}{(-1)^{n+2}} = (n+1)!$

$\Rightarrow f(x) = \sum_{n\geq 0} \frac{(n+1)!}{n!}(x+1)^n = \sum_{n\geq 0} (n+1)(x+1)^n.$

Notăm : $y = x+1$ și $a_n = n+1$ pt $\sum_{n\geq 0} a_n y^n$

$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n+2}{n+1} = 1 \Rightarrow R = \frac{1}{L} = 1$

pt $y = 1$: $\sum_{n\geq 0} (n+1)$. Folosim criteriul necesar.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (n+1) = +\infty \neq 0 \Rightarrow \sum_{n\geq 0} (n+1)$ divergentă

pt $y = -1$: $\sum_{n\geq 0} (n+1)(-1)^n$ folosim criteriul necesar

pt $n = 2K ; K \in \mathbb{N} \Rightarrow u_{2K} = (2K+1)(-1)^{2K} \xrightarrow{K\to\infty} \infty$

pt $n = 2K+1 ; K \in \mathbb{N} \Rightarrow u_{2K+1} = (2K+2)(-1)^{2K+1} \xrightarrow{K\to\infty} -\infty$ $\Rightarrow$

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$\Rightarrow \sum_{n\geq 0} u_n$ divergentă (Am notat $u_n = (-1)^n a_n$).

$\Rightarrow$ M. C în $y = (-1,1) \Rightarrow$ M. C în $X = (-2,0)$

4. $f(x) = \frac{1}{x^2+3x+2}$; $a = -4$

$$f(x) = \sum_{n\geq 0} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$f(x) = \frac{1}{x^2+3x+2} = \frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$\Rightarrow A(x+2) + B(x+1) = 1 \Rightarrow x(A+B) + 2A + B = 1 \Leftrightarrow$

$\Rightarrow \begin{cases} A+B=0 \\ 2A+B=1 \end{cases} \overset{\ominus}{\Rightarrow} A = 1 \Rightarrow B = -1 \Rightarrow$

$\Rightarrow f(x) = \frac{1}{x+1} - \frac{1}{x+2}$

$$f^{(n)}(x) = \frac{(-1)^n n!}{(x+1)^{n+1}} - \frac{(-1)^n n!}{(x+2)^{n+1}} \Rightarrow$$

$$\Rightarrow f^{(n)}(-4) = \frac{(-1)^n \cdot n!}{(-3)^{n+1}} - \frac{(-1)^n \cdot n!}{(-2)^{n+1}} = \frac{n!}{3^{n+1}} - \frac{n!}{2^{n+1}}$$

$$\Rightarrow f(x) = \sum_{n\geq 0} \frac{\cancel{n!}}{\cancel{n!}} \left(\frac{1}{3^{n+1}} - \frac{1}{2^{n+1}}\right)(x+4)^n = \sum_{n\geq 0} \left(\frac{1}{3^{n+1}} - \frac{1}{2^{n+1}}\right)(x+4)^n$$

Notăm: $x+4 = y$ și $a_n = \frac{1}{3^{n+1}} - \frac{1}{2^{n+1}}$ pt $\sum_{n\geq 0} a_n y^n$

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left| \frac{\frac{1}{3^{n+2}} - \frac{1}{2^{n+2}}}{\frac{1}{3^{n+1}} - \frac{1}{2^{n+1}}} \right| =$$

$$= \lim_{n\to\infty} \left| \frac{\frac{1}{2^{n+2}}\left[\left(\frac{2}{3}\right)^{n+2} - 1\right]}{\frac{1}{2^{n+1}}\left[\left(\frac{2}{3}\right)^{n+1} - 1\right]} \right| = \frac{1}{2} \Rightarrow R = \frac{1}{L} = 2$$

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pt $y=2$: $\sum_{n\geq 0}\left(\frac{1}{3^{n+1}}-\frac{1}{2^{n+1}}\right)2^n=\sum_{n\geq 0}\left(\frac{2^n}{3^{n+1}}-\frac{1}{2}\right)$

Folosim criteriul necesar (Notăm: $u_n=\frac{2^n}{3^{n+1}}-\frac{1}{2}$)

$\lim_{n\to\infty} u_n=\lim_{n\to\infty}\frac{2^n}{3^{n+1}}-\frac{1}{2}=-\frac{1}{2}\neq 0\Rightarrow \sum_{n\geq 0} u_n$ divergentă

pt $y=-2$: $\sum_{n\geq 0}\left(\frac{1}{3^{n+1}}-\frac{1}{2^{n+1}}\right)(-2)^n=\sum_{n\geq 0} u_n(-1)^n$.

Folosim criteriul Leibnitz:

Cunoaștem $\lim_{n\to\infty} u_n=0$ (1)

$$\frac{u_{n+1}}{u_n}=\frac{\frac{2^{n+1}}{3^{n+2}}-\frac{1}{2}}{\frac{2^n}{3^{n+1}}-\frac{1}{2}}=\frac{\frac{2^{n+2}-3^{n+2}}{2\cdot 3^{n+2}}}{\frac{2^{n+1}-3^{n+1}}{2\cdot 3^{n+1}}}=\frac{2^{n+2}-3^{n+2}}{3\cdot 2^{n+1}-3^{n+2}}<1$$

$\Rightarrow u_{n+1}<u_n\Rightarrow u_n \searrow$ (2)

(1)(2) $\Rightarrow \sum_{n\geq 0}(-1)^n u_n$ convergentă)

$\Rightarrow$ M.C în $y=[-2,2)\Rightarrow$ M.C în $x=[-6,-2)$.

• Folosind serile de puteri (și/sau seria Taylor), aproximați cu o eroare mai mică de 10^{-3} numerele:

1. $\sqrt[3]{65}$

$f(x)=\sqrt[3]{x}$; $f'(x)=\frac{1}{3}x^{-\frac{2}{3}}$; $f''(x)=-\frac{1\cdot 2}{3^2}\cdot x^{-\frac{5}{3}}$;

$f'''(x)=+\frac{2\cdot 5}{3^3}\cdot x^{-\frac{8}{3}}$; $f^{(IV)}(x)=-\frac{2\cdot 5\cdot 8}{3^4}x^{-\frac{11}{3}}$

$P(n)$: $f^{(n)}(x)=(-1)^n\cdot\frac{2\cdot 5\cdot\ldots\cdot(3n-4)}{3^n}\cdot x^{-\frac{3n-1}{3}}$; $\forall n\geq 1$

$P(1)$: $f'(x)=-1\cdot\frac{-1}{3}\cdot x^{-\frac{2}{3}}=\frac{1}{3}x^{-\frac{2}{3}}$ (Adevărat)

$P(n+1)$: $f^{(n+1)}(x)=(-1)^{n+1}\frac{2\cdot 5\cdot\ldots\cdot(3n-4)(3n-1)}{3^{n+1}}\cdot x^{-\frac{3n+2}{3}}$; $\forall n\geq 1$

$P(n)\rightarrow P(n+1)$: derivăm în $P(n)$

$\Rightarrow f^{(n+1)}(x)=(-1)^{n+1}\cdot\frac{2\cdot 5\cdot\ldots\cdot(3n-4)(3n-1)}{3^{n+1}}\cdot x^{\frac{-3n+2}{3}}$ (Adevărat)

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$\Rightarrow f^{(n)}(x) = (-1)^n \cdot \frac{2\cdot 5\cdot \ldots \cdot (3n-4)}{3^n} \cdot x^{-\frac{3n-1}{3}}$

Alegem $x = 65$ și $a = 64$.

$f(x) = \sum_{n\geq 0} \frac{f^{(n)}(a)}{n!}(x-a)^n = T_n(x) + R_n(x)$,

unde $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$; $x > c > a$

$R_n(x) = (-1)^{n+1} \cdot \frac{2\cdot 5\cdot \ldots \cdot (3n-1)}{3^{n+1}} \cdot c^{-\frac{3n+2}{3}} (65-64)^{n+1} =$

$= (-1)^{n+1} \cdot \frac{2\cdot 5\cdot \ldots \cdot (3n-1)}{3^{n+1}} \cdot c^{-\frac{3n+2}{3}}$

Avem: $64 < c < 65 \Rightarrow 64^{-\frac{3n+2}{3}} > c^{-\frac{3n+2}{3}} > 65^{-\frac{3n+2}{3}} \Rightarrow$

$\Rightarrow (-1)^{n+1} \cdot \frac{2\cdot 5\cdot \ldots \cdot (3n-1)}{3^{n+1}} \cdot 65^{-\frac{3n+2}{3}} < R_n(x) < (-1)^{n+1} \cdot \frac{2\cdot 5\cdot \ldots \cdot (3n-1)}{3^{n+1}} \cdot 64^{-\frac{3n+2}{3}}$

$\Rightarrow |R_n(x)| < \frac{2\cdot 5\cdot \ldots \cdot (3n-1)}{3^{n+1}} \cdot 4^{-(3n+2)} < \frac{1}{10^3}$

Dăm valori lui n:

$n = 1 \Rightarrow R_1(x) = \frac{2}{3^2} \cdot 4^{-5} = \frac{1}{3^2 \cdot 2^9} < \frac{1}{10^3}$

$\Rightarrow$ Aproximarea conține primii 2 termeni:

$\sqrt[3]{65} \cong 4 + \frac{1}{3} \cdot 4^{-2} = \frac{193}{48} \cong 4{,}0208$

2. $\sin(32^\circ)$

$f(x) = \sin x = \sum_{n\geq 0} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!}$

$32^\circ = \frac{8\pi}{45} \Rightarrow f(32^\circ) = \sum_{n\geq 1} (-1)^n \cdot \left(\frac{8\pi}{45}\right)^{2n+1} \cdot \frac{1}{(2n+1)!}$

Notăm: $a_n = \left(\frac{8\pi}{45}\right)^{2n+1} \cdot \frac{1}{(2n+1)!}$ și folosim criteriul Leibnitz:

$\frac{a_{n+1}}{a_n} = \left(\frac{8\pi}{45}\right)^{2n+3} \cdot \frac{1}{(2n+3)!} \cdot \left(\frac{45}{8\pi}\right)^{2n+1} \cdot (2n+1)! = \frac{\left(\frac{8\pi}{45}\right)^2}{(2n+2)(2n+3)}$

$< 1 \Rightarrow a_{n+1} < a_n \Rightarrow a_n \searrow 0$ (1)

II.

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$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \underbrace{\left(\frac{8\sqrt{11}}{45}\right)^{2n+1}}_{\to 0} \cdot \underbrace{\frac{1}{(2n+1)!}}_{\to 0} = 0$ (2)

(1)(2) $\Rightarrow \sum_{n\geq 0} (-1)^n a_n$ convergentă $\Rightarrow$ seria are sumă.

Notăm: S = suma seriei ; S_n = șirul sumelor parțiale.

Trebuie să determinăm cel mai mic $n \in \mathbb{N}$ a.î.

$|S - S_n| < a_{n+1} < \frac{1}{10^4}$; $\frac{8\sqrt{11}}{45} \approx 0{,}55$

$a_1 = \left(\frac{8\sqrt{11}}{45}\right)^3 \cdot \frac{1}{3!} = \frac{11^3}{2^4 \cdot 10^3 \cdot 3}$; $a_0 = \frac{8\sqrt{11}}{45} \cdot 1 \approx 0{,}55$;

$a_2 = \left(\frac{8\sqrt{11}}{45}\right)^5 \cdot \frac{1}{5!} = \frac{11^5}{2^5 \cdot 10^5 \cdot 5!}$; $a_3 = \left(\frac{8\sqrt{11}}{45}\right)^7 \cdot \frac{1}{7!} < 10^{-4}$

$= 0{,}000003\ldots$

$\Rightarrow n+1 = 3 \Rightarrow n = 2$

$\Rightarrow S_2 = a_0 - a_1 + a_2 = 0{,}5500 - 0{,}0277 + 0{,}0004 =$

$= 0{,}5227$

$\Rightarrow -a_3 < S - S_2 < a_3 \mid + S$

$S_2 - a_3 < S < S_2 + a_3$

$0{,}5227 - 0{,}000003 < S < 0{,}5227 + 0{,}000003$

$0{,}5227\ldots < S < 0{,}5227\ldots$

$\Rightarrow S \cong 0{,}5227$ (cu 4 zecimale)

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